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PART3:

Feature Matrix: $X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}$

Target vector: $Y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$

Initial Parameters: $m = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$

Bias: $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Initial Prediction:

Prediction: $Xm = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix} = \begin{bmatrix} 5 \\ 16 \end{bmatrix}$

Add bias: $\hat{y} = Xm + b = \begin{bmatrix} 5 \\ 16 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$

$$\hat{y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

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Derive the Gradients:

Mean Squared Error: $J(m,b) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$

where, $\hat{y} = Xm + b$

Error vector, $e = \hat{y} - y$

therefore, $J = \frac{1}{n}e^T e$

Gradient with respect to m $\frac{\partial J}{\partial b} = \frac{2}{n}e$

here, $n = 2$, therefore $\frac{2}{n} = 1$

so, $\nabla_m = X^T e$

$\nabla_b = e$

where X^T is the transpose of X

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We set learning rate $\alpha = 0.01$

__Iteration 1__ :

Error : $e = \hat{y} - y = \begin{bmatrix} 6 \\ 17 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

Gradient of m : $X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$

$$\nabla_m = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

Gradient of bias : $\nabla_b = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$

update m : $m_1 = m - \alpha \nabla_m$

$$m_1 = \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 0.01 \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix}$$

update bias : $b_1 = b - \alpha \nabla_b$

$$b_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.01 \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix}$$

Cost function : $J_1 = \frac{1}{2}(1^2 + 11^2) = 61$

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Iteration 2 :

Prediction : $Xm_1 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} = \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix}$

Add bias : $\hat{y} = Xm + b = \begin{bmatrix} 1.16 \\ 2.9 \end{bmatrix} + \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix}$

Error : $e = \hat{y} - y = \begin{bmatrix} 2.15 \\ 3.79 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$

Gradient of m : $\nabla_m = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix} = \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$

Gradient of bias : $\nabla_b = e = \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$

update m : $m_2 = m_1 - \alpha \nabla_m = \begin{bmatrix} -1.45 \\ 0.87 \end{bmatrix} - 0.01 \begin{bmatrix} -11.69 \\ -30.65 \end{bmatrix}$

$= \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix}$

update b : $b_2 = b_1 - \alpha \nabla_b = \begin{bmatrix} 0.99 \\ 0.89 \end{bmatrix} - 0.01 \begin{bmatrix} -2.85 \\ -2.21 \end{bmatrix}$

$= \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix}$

cost Function : $J_2 = \frac{1}{2}((-2.85)^2 + (-2.21)^2) = 6.503$

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Iteration 3:

Prediction: $Xm_2 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} = \begin{bmatrix} 2.1964 \\ 6.4346 \end{bmatrix}$

Add bias: $\hat{y} = Xm + b = \begin{bmatrix} 2.1964 \\ 6.4346 \end{bmatrix} + \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} = \begin{bmatrix} 3.2149 \\ 7.3467 \end{bmatrix}$

Error: $e = \hat{y} - y = \begin{bmatrix} 3.2149 \\ 7.3467 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -1.7851 \\ 1.3467 \end{bmatrix}$

Gradients: $\nabla_m = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.7851 \\ 1.3467 \end{bmatrix} = \begin{bmatrix} 3.6017 \\ 8.1117 \end{bmatrix}$

$$\nabla_b = e = \begin{bmatrix} -1.7851 \\ 1.3467 \end{bmatrix}$$

Update: $m_3 = m_2 - \alpha \nabla_m = \begin{bmatrix} -1.3331 \\ 1.1765 \end{bmatrix} - 0.01 \begin{bmatrix} 3.6017 \\ 8.1117 \end{bmatrix} =$

$$m_3 = \begin{bmatrix} -1.3691 \\ 1.0954 \end{bmatrix}$$

$$b_3 = b_2 - \alpha \nabla_b = \begin{bmatrix} 1.0185 \\ 0.9121 \end{bmatrix} - 0.01 \begin{bmatrix} -1.7851 \\ 1.3467 \end{bmatrix} =$$

$$b_3 = \begin{bmatrix} 1.0364 \\ 0.8986 \end{bmatrix}$$

Cost Function: $J_3 = \frac{1}{2}\left((-1.7851)^2 + (1.3467)^2\right) = 2.5$

Iteration 4:

Prediction : $X m_3 = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3691 \\ 1.0954 \end{bmatrix} = \begin{bmatrix} 1.9171 \\ 5.4766 \end{bmatrix}$

Add bias : $\hat{y} = X m_3 + b_3 = \begin{bmatrix} 1.9171 \\ 5.4766 \end{bmatrix} + \begin{bmatrix} 1.0364 \\ 0.8986 \end{bmatrix} = \begin{bmatrix} 2.9535 \\ 6.3752 \end{bmatrix}$

Error : $e = \hat{y} - y = \begin{bmatrix} 2.9535 \\ 6.3752 \end{bmatrix} - \begin{bmatrix} 5 \\ 6 \end{bmatrix} = \begin{bmatrix} -2.0465 \\ 0.3752 \end{bmatrix}$

Gradients : $\nabla_m = X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.0465 \\ 0.3752 \end{bmatrix} = \begin{bmatrix} -0.5457 \\ -2.3860 \end{bmatrix}$

$$\nabla_b = e = \begin{bmatrix} -2.0465 \\ 0.3752 \end{bmatrix}$$

updates :

$$m_4 = m_3 - \alpha \nabla_m = \begin{bmatrix} -1.3691 \\ 1.0954 \end{bmatrix} - 0.01 \begin{bmatrix} -0.5457 \\ -2.3860 \end{bmatrix} = \begin{bmatrix} -1.3636 \\ 1.1193 \end{bmatrix}$$

$$b_4 = b_3 - \alpha \nabla_b = \begin{bmatrix} 1.0364 \\ 0.8986 \end{bmatrix} - 0.01 \begin{bmatrix} -2.0465 \\ 0.3752 \end{bmatrix} = \begin{bmatrix} 1.0569 \\ 0.8948 \end{bmatrix}$$

Cost Function : $J_4 = \frac{1}{2}\left((-2.0465)^2 + (0.3752)^2\right)$

$$= 2.1644$$

Trend observation

Through all iterations, the cost function went down very rapidly from 61.0 to 2.16, however, the decreasing speed became slower at each stage(from 6.5 to 2.5 to 2.16). This demonstrates that gradient descent worked correctly(large, fast improvement early on when far from the minimum, followed by smaller refinements as the model gets closer to the optimal parameters). The bias b increased steadily and smoothly during all the iterations, which means it tended to reach its optimal value straightly. The behaviour of m1 and m2 was different. While m1 converged almost immediately, m2 first reached a larger than optimal value at iteration 1, then oscillated and started converging slowly after this point. This indicates that the learning rate is adequate for these parameters but is not tuned well enough.
The values are still showing convergence and not divergence trends as the error keeps reducing each time.